

Zhongze LUO

☎ (+86)136-0797-9355 | ✉ luozhongze0928@foxmail.com | 🔗 <https://luozhongze.github.io>

Education

The Chinese University of Hong Kong, Shenzhen ([T-Lab](#) [\[link\]](#)) (Expected) 2025.09—2027.06
M.Phil. of Computer and Information Engineering GPA: 3.87/4.00 Ranking: 2/54 (Top 5%)

Northeast Forestry University ([Project 211](#)) 2021.09—2025.06
B.Eng of Communication Engineering GPA: 88.08/100 Ranking: 14/73 (Top 20%)

Research interests

Multimodal Understanding, LLM Reasoning and Evaluation, Knowledge Graph.

Projects

Off-Distribution Voices

[Under Review](#)

Constructed a register-based jailbreak framework by conditioning creative-writing prompts on passages from 12 real fanfiction subgenres and embedding harmful requests as narrative climaxes, without per-target prompt optimization or attacker-side LLM generation. Conducted large-scale evaluations on 8 aligned LLMs over HarmBench and JailbreakBench using a four-judge ensemble, and performed factorial ablations to isolate the contribution of linguistic register from prompt length and structure. Further proposed SAGA-A4, a static four-turn extension that achieved a mean ASR of 0.924 and exposed the limitations of template-oriented safety defenses.

Multi-Physics

[ICASSP 2026](#)

Built Multi-Physics, a Chinese multimodal physics-reasoning benchmark with 1,412 image-associated multiple-choice questions, covering 11 high-school physics subjects and 5 difficulty levels. Designed a dual evaluation protocol combining final-answer accuracy with step-by-step reasoning integrity, and benchmarked 20 MLLMs under text-only and image-text settings. Conducted controlled analyses of model performance across subjects, difficulty levels, and input modalities, revealing gaps in visual grounding and scientific reasoning; released the dataset and evaluation code.

KG2QA

[ICASSP 2026](#)

Developed KG2QA, an end-to-end question-answering system for communication standards by constructing a 6,587-pair QA dataset from ITU-T recommendations and fine-tuning Qwen2.5-7B-Instruct. Built a domain knowledge graph with 13,906 entities and 13,524 relations through LLM-assisted triple extraction guided by a custom ontology, then integrated it into a KG-RAG pipeline for knowledge retrieval and grounded answer generation. Compared against base LLM and RAG baselines using automatic metrics and DeepSeek-V3-as-a-judge evaluation; deployed the resulting system through a web platform and API.

Skills

LLM Training: LLaMA-Factory, SFT, LoRA & QLoRA, and RL-based alignment, PPO, DPO, GRPO.

RAG & Knowledge Graph: RAG & KG, Neo4j, Cypher, ontology design, triple extraction, graph-enhanced QA.

Evaluation: Benchmark construction, LLM-as-a-Judge, CoT evaluation; metrics including ASR, BLEU-4.

Engineering: Python, PyTorch, model training & inference, API deployment; target detection and edge-computing.

Language: IELTS Academic 6.5 (Listening 7.0, Reading 7.5, Writing 6.0, Speaking 6.0).

Papers

- **Luo, Z.**, Shi, R., Yin, Z., Liu, H., Wan, W., & Tang, X. (2026). Off-Distribution Voices: Fanfiction Subgenres as Universal Vernacular Jailbreaks for Aligned LLMs. arXiv preprint arXiv:2606.04483. [\[link\]](#)
- **Luo, Z.**, Yin, Z., Guo, Y., Wang, Z., Zhu, J., Tang, X. (2026). Multi-Physics: A Comprehensive Benchmark for Multimodal LLMs Reasoning on Chinese Multi-subject Physics Problems. ICASSP 2026. [\[link\]](#)
- **Luo, Z.**, Wan, W., Zhang, T., Wang, D., Tang, X. (2026). KG2QA: Knowledge Graph-enhanced Retrieval-augmented Generation for Communication Standards Question Answering. ICASSP 2026. [\[link\]](#)
- Sun, J., & **Luo, Z.** (2025). ForPKG: A Framework for Constructing Forestry Policy Knowledge Graph and Application Analysis. In 2025 International Joint Conference on Neural Networks (IJCNN). [\[link\]](#)
- Sun, J., **Luo, Z.**, & Li, Y. (2025). A compliance checking framework based on retrieval augmented generation. In Proceedings of the 31st International Conference on Computational Linguistics (COLING). [\[link\]](#)

- Zheng, Q., **Luo, Z.**, Guo, M., Wang, X., Wu, R., Meng, Q., & Dong, G. (2025). HGO-YOLO: advancing anomaly behavior detection with hierarchical features and lightweight optimized detection. Journal of Real-Time Image Processing, 22(4), 1-15. [\[link\]](#)
- **Luo, Z.**, Jia, S., Niu, H., Zhao, Y., Zeng, X., & Dong, G. (2024). Elderly fall detection algorithm based on improved yolov5s. Information Technology and Control, 53(2), 601-618. [\[link\]](#)
- Sun, J., Dai, C., **Luo, Z.**, Chang, Y., & Li, Y. (2024). LawLuo: A Multi-Agent Collaborative Framework for Multi-Round Chinese Legal Consultation. arXiv preprint arXiv:2407.16252. [\[link\]](#)

Honors

Outstanding Undergraduate Graduation Thesis of Northeast Forestry University (3%)	2025.06
China College Student Computer Design Competition - Third Prize	2024.08
National College Students Mathematical Contest in Modeling - Provincial First Prize	2022.11
University scholarship - First Prize	2022.03